UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER SECURITY
COURSE CODE	:	CSC662
EXAMINATION	:	OCTOBER 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC662 mock paper (October 2026). Computed answers show working; labels are computed / convention unverified / placeholder.

2. Theory questions are placeholders - the theory module has not been built.

3. Part A = 10 x (2 + 6); Part B = 2 x (5 + 5); total = 100 marks.

4. Computed: 6 slots / 28 marks. Placeholders: 18 slots / 72 marks. Questions and answers are for local study use only.

5. See the review-pack checklist for the Playfair convention and the two OTP alphabet variants.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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CSC662 - MARKING SCHEME (MOCK OCTOBER 2026)

Confidence labels: computed / convention unverified / placeholder. No official marking scheme exists for the real CSC662 papers; this scheme is generated for the mock.

Part A: 10 x (2 + 6) = 80 marks. Part B: 2 x (5 + 5) = 20 marks. Total: 100 marks.

PART A Q2(a) - 2 marks - computed - matches lecture slides 5-9 (Caesar shift 3 examples)

Question:

Using Caesar Cipher technique, decrypt the following ciphertext with a shift of 7 (replace each letter with the one 7 positions to the left).
KLCPJLYLABYULKHUKDPWLKVUL

Answer: DEVICERETURNEDANDWIPEDONE

Working:

```
K (10) -7 mod 26 -> D
L (11) -7 mod 26 -> E
C ( 2) -7 mod 26 -> V
P (15) -7 mod 26 -> I
J ( 9) -7 mod 26 -> C
L (11) -7 mod 26 -> E
Y (24) -7 mod 26 -> R
L (11) -7 mod 26 -> E
A ( 0) -7 mod 26 -> T
B ( 1) -7 mod 26 -> U
Y (24) -7 mod 26 -> R
U (20) -7 mod 26 -> N
L (11) -7 mod 26 -> E
K (10) -7 mod 26 -> D
H ( 7) -7 mod 26 -> A
U (20) -7 mod 26 -> N
K (10) -7 mod 26 -> D
D ( 3) -7 mod 26 -> W
P (15) -7 mod 26 -> I
W (22) -7 mod 26 -> P
L (11) -7 mod 26 -> E
K (10) -7 mod 26 -> D
V (21) -7 mod 26 -> O
U (20) -7 mod 26 -> N
L (11) -7 mod 26 -> E
```

Suggested marks: 1 mark for identifying the cipher/shift direction; 1 mark for the complete plaintext.
Confidence: computed - matches lecture slides 5-9 (Caesar shift 3 examples).

PART A Q2(b) - 6 marks - convention unverified - no lecture slide covers Playfair; I/J merge + X padding still to be confirmed

Question:

Encrypt the message below using Playfair encryption with the key REDFOX. Use a 5 x 5 grid, merge I and J, and use X as the filler between repeated letters and at the end if needed.
REVIEW ACCESS PERMISSIONS

Answer: EDEPDVBGAFNCTQDEHKNCPLRTNC

Working:

```
Grid (I/J merged):
  R E D F O
  X A B C G
  H I K L M
  N P Q S T
  U V W Y Z
Padded pairs: RE  VI  EW  AC  CE  SX  SP  ER  MI  SX  SI  ON  SX
RE -> ED   (same row 1: shift right)
VI -> EP   (same column 2: shift down)
EW -> DV   (rectangle (1,2)-(5,3): swap columns)
```

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```
AC -> BG   (same row 2: shift right)
CE -> AF   (rectangle (2,4)-(1,2): swap columns)
SX -> NC   (rectangle (4,4)-(2,1): swap columns)
SP -> TQ   (same row 4: shift right)
ER -> DE   (same row 1: shift right)
MI -> HK   (same row 3: shift right)
SX -> NC   (rectangle (4,4)-(2,1): swap columns)
SI -> PL   (rectangle (4,4)-(3,2): swap columns)
ON -> RT   (rectangle (1,5)-(4,1): swap columns)
SX -> NC   (rectangle (4,4)-(2,1): swap columns)
Ciphertext: EDEPDVBGAFNCTQDEHKNCPLRTNC
```

Suggested marks: 1 mark for the correct 5x5 grid; 1 mark for correct pairing/padding; 4 marks for the ciphertext pairs.

Confidence: convention unverified - no lecture slide covers Playfair; I/J merge + X padding still to be confirmed.

PART A Q3(b) - 6 marks - computed (consistency machine-verified)

Question:

Assume a computer system with the accounts: Ali, Yati, Umi, Root. Ali can read, write X-1; Yati can read X-1; Umi can execute REPORT; Root can read, write, execute DATA.

Based on the scenario above, write the access control matrix (ACM), the access control lists (ACL) and the capability lists (CL), and draw the diagram.

Answer:

Subject \ Object	X-1	REPORT	DATA
Ali	read, write	-	-
Yati	read	-	-
Umi	-	execute	-
Root	-	-	execute, read, write

ACLs (per object):

X-1: Ali: read, write; Yati: read
REPORT: Umi: execute
DATA: Root: execute, read, write

Capability lists (per subject):

Ali: X-1: read, write
Yati: X-1: read
Umi: REPORT: execute
Root: DATA: execute, read, write

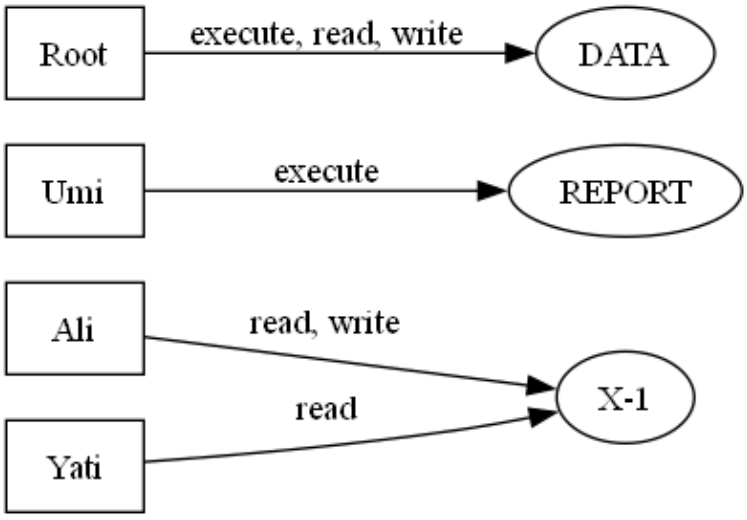


Figure: access-control diagram (answer).

Suggested marks: 2 matrix, 2 ACL, 1 capability list, 1 diagram.

Confidence: computed - the matrix, ACLs and capability lists rebuild each other without loss.

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PART A Q4(a) - 2 marks - computed - matches lecture slides 5-9 (Caesar shift 3 examples)

Question:

Using Caesar Cipher technique, decrypt the following ciphertext with a shift of 7 (replace each letter with the one 7 positions to the left).
AYHPULTWSVFLLZVUZLJBYPAF

Answer: TRAINEMPLOYEESONSECURITY

Working:

```
A ( 0) -7 mod 26 -> T
Y (24) -7 mod 26 -> R
H ( 7) -7 mod 26 -> A
P (15) -7 mod 26 -> I
U (20) -7 mod 26 -> N
L (11) -7 mod 26 -> E
T (19) -7 mod 26 -> M
W (22) -7 mod 26 -> P
S (18) -7 mod 26 -> L
V (21) -7 mod 26 -> O
F ( 5) -7 mod 26 -> Y
L (11) -7 mod 26 -> E
L (11) -7 mod 26 -> E
Z (25) -7 mod 26 -> S
V (21) -7 mod 26 -> O
U (20) -7 mod 26 -> N
Z (25) -7 mod 26 -> S
L (11) -7 mod 26 -> E
J ( 9) -7 mod 26 -> C
B ( 1) -7 mod 26 -> U
Y (24) -7 mod 26 -> R
P (15) -7 mod 26 -> I
A ( 0) -7 mod 26 -> T
F ( 5) -7 mod 26 -> Y
```

Suggested marks: 1 mark for identifying the cipher/shift direction; 1 mark for the complete plaintext.
Confidence: computed - matches lecture slides 5-9 (Caesar shift 3 examples).

PART A Q4(b) - 6 marks - convention unverified - no lecture slide covers Playfair; I/J merge + X padding still to be confirmed

Question:

Encrypt the message below using Playfair encryption with the key REDFOX. Use a 5 x 5 grid, merge I and J, and use X as the filler between repeated letters and at the end if needed.
CONTAINER SECURITY CHECKS

Answer: GFPNIPPRFNFARXMPFLIRBLNC

Working:

```
Grid (I/J merged):
  R E D F O
  X A B C G
  H I K L M
  N P Q S T
  U V W Y Z
Padded pairs: CO  NT  AI  NE  RS  EC  UR  IT  YC  HE  CK  SX
CO -> GF   (rectangle (2,4)-(1,5): swap columns)
NT -> PN   (same row 4: shift right)
AI -> IP   (same column 2: shift down)
NE -> PR   (rectangle (4,1)-(1,2): swap columns)
RS -> FN   (rectangle (1,1)-(4,4): swap columns)
EC -> FA   (rectangle (1,2)-(2,4): swap columns)
UR -> RX   (same column 1: shift down)
IT -> MP   (rectangle (3,2)-(4,5): swap columns)
YC -> FL   (same column 4: shift down)
HE -> IR   (rectangle (3,1)-(1,2): swap columns)
CK -> BL   (rectangle (2,4)-(3,3): swap columns)
SX -> NC   (rectangle (4,4)-(2,1): swap columns)
Ciphertext: GFPNIPPRFNFARXMPFLIRBLNC
```

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Suggested marks: 1 mark for the correct 5x5 grid; 1 mark for correct pairing/padding; 4 marks for the ciphertext pairs.

Confidence: convention unverified - no lecture slide covers Playfair; I/J merge + X padding still to be confirmed.

PART A Q5(b) - 6 marks - computed - 3 x 4 fill-rows/read-columns convention verified against lecture slides 20-21 and 24

Question:

Decrypt the following ciphertext using columnar transposition with a 3 x 4 matrix (3 rows, 4 columns). Put the ciphertext into the columns, read the plaintext out of the rows, and ignore the filler X at the end. The key is the column size, 4.
LNIOATGUXIDX

Answer: LOGINAUDIT

Working:

```
Numeric key: 4 columns; grid 3 rows x 4 columns; read order: col 1, col 2, col 3, col 4
Grid: 4 columns x 3 rows (12 cells; ciphertext 12 chars; transport pad 0)
Ciphertext split in read order:
  col 1: LNI
  col 2: OAT
  col 3: GUX
  col 4: IDX
Reconstructed rows:
  row 1: LOGI
  row 2: NAUD
  row 3: ITXX
Row-wise: LOGINAUDITXX
Remove 2 trailing pad 'X': LOGINAUDIT
```

Suggested marks: 1 mark for the key column order; 2 marks for reconstructing the columns; 2 marks for the row-wise plaintext; 1 mark for stripping the padding.

Confidence: computed - 3 x 4 fill-rows/read-columns convention verified against lecture slides 20-21 and 24.

PLACEHOLDER SLOTS (18 slots, 72 marks)

Theory questions are placeholders. The theory module has not been built, so no model answers are provided for these slots.

Slot	Topic	Marks	Task	Status
A Q1(a)	attacks/malware	2	short objective	placeholder
A Q1(b)	attacks/malware	6	define/explain	placeholder
A Q3(a)	access control models	2	short objective	placeholder
A Q5(a)	authentication/biometrics	2	short objective	placeholder
A Q6(a)	software/OS security	2	short objective	placeholder
A Q6(b)	software/OS security	6	define/explain	placeholder
A Q7(a)	network security	2	short objective	placeholder
A Q7(b)	network security	6	define/explain	placeholder
A Q8(a)	trusted computing	2	short objective	placeholder
A Q8(b)	trusted computing	6	define/explain	placeholder
A Q9(a)	authentication/biometrics	2	short objective	placeholder
A Q9(b)	authentication/biometrics	6	define/explain	placeholder
A Q10(a)	attacks/malware	2	short objective	placeholder
A Q10(b)	attacks/malware	6	define/explain	placeholder
B Q1(a)	attacks/malware	5	scenario/justify	placeholder
B Q1(b)	attacks/malware	5	scenario/justify	placeholder
B Q2(a)	authentication/biometrics	5	scenario/justify	placeholder
B Q2(b)	authentication/biometrics	5	scenario/justify	placeholder

End of marking scheme.

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